

CURRICULUM VITAE

Arabinda Haldar, Department of Physics, IIT Hyderabad
Hyderabad, October 30, 2025

ADDRESS

Arabinda Haldar

Associate Professor

Department of Physics (Office: PH406)

IIT Hyderabad, Kandi, Sangareddy 502284

Telangana, India

Email: arabinda@phy.iith.ac.in

Phone: +91-2301-6717, Mobile: +91-9000916586

Research group webpage: <https://iith.ac.in/~arabinda/>

Google scholar page: <https://scholar.google.co.in/citations?user=rGcjZqkAAAAJ&hl=en>

Scopus page: <https://www.scopus.com/authid/detail.uri?authorId=25633944300>

ORCID page: <https://orcid.org/0000-0002-0490-9719>

RESEARCH INTERESTS

Magnonics, Spin dynamics, Spin-orbitronics, Magnetic thin films and nanostructures, Ferromagnetic resonance, Brillouin light scattering spectro-microscopy, Skyrmions, Micromagnetics, Nanofabrication, Nanomagnetic devices.

CAREER PROFILE

- **14+ years of post-Ph. D. research experience** and one year of industrial research concurrently with Ph.D.
- Average **teaching feedback ~ 4.1 over 9 years**. Taught 55 credits of theory courses over 9 yrs.
- Co-authored **61 research papers, 4 granted+6 filed patents** since joining IIT Hyderabad in Aug 2016.
- Total publications = 87 (including 2 book chapters, 4 invited reviews), Granted US patents: 2, Granted Indian patents: 4, Filed Indian patents: 6
- Google scholar: h-index = 20 and citations = 1782; Scopus: h-index = 18 and citations = 1320
- **Total of eight sponsored projects of worth ~ 3 Cr.** Completed projects = 6. ECR project completed with a 'very good' rating.

AWARDS/RECOGNITIONS (since joining IITH)

- Selected as a member of the '**Editorial Board**' of the '**Scientific Reports**' journal (Q1, IF: 3.9) (Nature Publishing Group), since March 2025.
- Visit to NIMS, Japan, for two weeks as a **Guest Researcher**, November 2024.
- Recipient of the highly competitive and prestigious '**Research Excellence Award**' from IIT Hyderabad, 2024.
- Session Chair for ICMAGMA 2023, Hyderabad, India Dec 4-6, 2023.
- **Session Chair for INTERMAG 2023**, Sendai, Japan, May 15-19, 2023.
- Session Chair for ICONN 2023, SRM Institute of Science and Technology, Chennai, Mar 27-29, 2023.
- Session Chair for 4th International Conference on Current Trends in Materials Science and Engineering (CTMSE 2022), Institute of Engineering & Management, Kolkata, 28 – 30 July 2022.

- **'Early Career Editorial Board'** member for *Materials Today Electronics* (Elsevier) journal (Q1, IF: 7.4), 2022.
- **'Review Editor'** on the Editorial Board of Quantum Materials (specialty section of *Frontiers in Materials*) (Q2, IF: 2.9), 2022.
- Recognized as an **'IEEE Senior Member'**, IEEE, 2022.
- Recipient of prestigious **Young Scientist Research Award** (YSRA), BRNS, 2021.
- **Invited speaker** at the International Conference on Magnetism (**INTERMAG 2021**), April 26-30, 2021 (held online due to COVID, due to be held in Lyon, France).
- Invited by J. of Appl. Phys. (AIP) to write a **perspective** on emerging reconfigurable magnonic metamaterials, 2020.
- **Outstanding Reviewer** 2019 for J. of Phys. D: Appl. Phys, by IOP
- Invited as a **Guest Editor** for the 'Frontiers in Physics' journal, 2019.
- Recipient of **Early Career Research Award**, SERB (now ANRF), 2018.
- Recipient of prestigious **Ramanujan Fellowship**, 2017.
- Reviewer of various esteemed high-impact journals, including PRL, PRB, PRApplied, Nature Communications, Nano Letters, and Nanoscale.

JOB EXPERIENCE

- **Associate Professor: Oct 18, 2021 – Present (4 years)**
Indian Institute of Technology Hyderabad, India
- **Guest Researcher: (Nov 24 – Dec 6, 2024)**
NIMS, Japan
- **Assistant Professor: August 1, 2016 – Oct 17, 2021 (5 years, 2 months)**
Indian Institute of Technology Hyderabad, India
- **Research Fellow: June 3, 2014 – July 21, 2016 (2 years)**
Information Storage Materials Lab., Dept. of Elec. and Comp. Engg., National University of Singapore, Singapore
- **Scientist-C: Mar 13, 2013 – May 30, 2014 (1 year, 2 months)**
Thematic Unit for Nanodevice and Technology, S. N. Bose National Centre for Basic Sciences, Kolkata, India
- **Postdoctoral Research Associate: Jan 24, 2011 – Mar 8, 2013 (2 years, 1 month)**
Department of Physics, Colorado State University, Fort Collins, USA.

EDUCATION

- **M.Sc. & Ph.D Dual Degree**, Physics, CPI: 8.62/10, Indian Institute of Technology Bombay, Mumbai, India, 2005-2011.
Ph.D : Magnetism in functional magnetic materials (Supervisor: Prof. K. G. Suresh)
- **B.Sc with Physics honors**, 1st class, Ramakrishna Mission Vidyamandira, Belur, University of Calcutta, India, 2002-2005.

TEACHING AT IIT HYDERABAD:

- 🚦 No of different course types: Core = 5, Lab = 4 and Elective = 6
- 🚦 Taught 55 credits of theory courses over a period of 9 years
- 🚦 Avg. course feedback: ~ 4.1 over a period of 9 years

S. No.	Academic Period	Course Code	Course Name	Credit	Total Students	Score
1	JAN-MAY 2025	EP3200	Solid State Physics	3	29	4.37
2	JAN-MAY 2025	PH56230	Spin Logic Systems.	2	15	4.55
	JAN-MAY 2025	PH5800	Spin Logic Systems		8	3.4
3	JUL-NOV 2024	EP5201	Core-EP Lab-5	2	25	4.18
	JUL-NOV 2024	EP5201	Core-EP Lab-5		5	4.33
4	JUL-NOV 2024	PH5310	Introduction to Nanomagnetism with Lab	3	9	4.8
	JUL-NOV 2024	EP4510	Introduction to Nanomagnetism with Lab		10	4.6
5	JAN-MAY 2024	EP1031	Physics Lab	2	123	4.02
	JAN-MAY 2024	EP1031	Physics Lab		44	4.13
	JAN-MAY 2024	EP1031	Physics Lab		66	3.68
6	JAN-MAY 2024	PH5800	Spin logic systems	2	6	4.96
7	JUL23-NOV23	PH5310	Introduction to Nanomagnetism with Lab	3	22	4.68
	JUL23-NOV23	EP4510	Introduction to Nanomagnetism with Lab		2	5
8	JUL23-NOV23	IS5100	Introduction to Sensors (3 credit: shared)	1	11	4.78
9	JAN23-MAY23	EP1031	Physics Lab	2	158	3.83
10	JAN23-MAY23	PH5350	Experimental Techniques	3	9	4.3
11	OCT22-FEB23_UG I SEM (Mar-Jul 23)	EP1031	Physics Lab	2	146	
12	AUG22-NOV22	PH5310	Introduction to Nanomagnetism with Lab	3	18	4.35
13	NOV21-MAR22_UG I SEM	EP1031	Physics Lab	2	125	3.6
14	MAR22-JUL22_UG IISEM	EP1031	Physics Lab	2	194	3.91
15	AUG21 - NOV21	EP4518	Introduction to Nanomagnetism	2	7	3.13
	AUG21 - NOV21	PH7318	Introduction to Nanomagnetism		25	4.28
16	AUG21 - NOV21	PH6292	Experimental Techniques-2	2	22	4.5
	AUG21 - NOV21	EP3498	Experimental Techniques-2		1	4
17	NOV20-FEB21_UG I SEM	EP3367	Experimental Techniques	1	21	3.45
	NOV20-FEB21_UG I SEM	PH6167	Experimental Techniques		25	4.22

18	NOV20-FEB21_UG I SEM	PH5311	Physics Lab (MSc2)	2	18	4.09
19	NOV20-FEB21_UG I SEM	PH4168	Spintronics	2	1	4.29
	NOV20-FEB21_UG I SEM	PH6168	Spintronics		18	4.68
20	MAR21-JUN21 UG IISEM	PH5288	Digital Electronics	1	54	4.2
21	JAN21-APR21	PH5311	Physics Lab (MSc2)	2	18	4.09
22	JAN21-APR21	PH5288	Digital Electronics	1	54	4.2
23	JAN21-APR21	PH4168	Spintronics	2	1	4.29
	JAN21-APR21	PH6168	Spintronics		18	4.68
24	AUG20 - NOV20	EP3367	Experimental Techniques	1	21	3.45
	AUG20 - NOV20	PH6167	Experimental Techniques		25	4.22
25	AUG20 - NOV20	EP3498	Experimental Techniques-2	2	2	4.63
	AUG20 - NOV20	PH6292	Experimental Techniques-2		14	4.65
26	JAN20 - APR20	PH5211	MSc Lab - II	2	18	3.68
27	JAN20 - APR20	PH6168	Spintronics	2	10	3.81
28	JAN20 - APR20	PH5288	Digital Electronics	1	45	3.88
29	JAN20 - APR20	EP2041	Physics Lab EP-2	2	20	4.23
30	AUG19 - NOV19	EP4368	Advanced Functional Materials	2	8	3.53
	AUG19 - NOV19	PH6338	Advanced Functional Materials		18	4.19
31	AUG19 - NOV19	PH6327	Nuclear Physics	1	19	3.57
	AUG19 - NOV19	EP3537	Nuclear Physics		9	3.77
32	AUG19 - NOV19	PH5311	Physics Lab (MSc2)	2	18	3.68
33	JAN19 - APR19	PH5288	Digital Electronics	1	20	4.06
	JAN18 - APR18	PH5288	Digital Electronics		21	3.79
34	JAN19 - APR19	PH6168	Spintronics	2	12	4.48
	JAN18 - APR18	PH6168	Spintronics		6	4.68
35	JAN18 - APR18	PH5211	Physics Lab - II	2	17	3.93
36	AUG18 - NOV18	EP1031	Physics Lab	2	198	3.99
37	AUG18 - NOV18	PH6338	Advanced Functional Materials	2	15	4
38	AUG18 - NOV18	PH6292	Experimental Techniques-2	2	12	4.08
39	JAN18 - APR18	PH5288	Digital Electronics	1	21	3.79
40	JAN18 - APR18	PH5211	Physics Lab - II	2	17	3.93
41	JAN18 - APR18	PH6168	Spintronics	2	6	4.68
42	AUG17 - NOV17	PH4071	EP5-Lab	2	9	3
43	AUG17 - NOV17	PH1031	B. Tech-Lab	2	181	3.58
44	AUG17 - NOV17	PH5311	MSc-Lab3	2	13	4.09
45	AUG17 - NOV17	PH6338	Advanced Functional Materials	2	8	4.75
46	JAN17-APR17	PH2817	Modern Physics	1	51	3.78
47	JAN17-APR17	PH4168	Spintronics	2	4	4
	JAN17-APR17	PH6168	Spintronics		12	4.43
48	Aug-Nov	PH1031	Physics lab	2	164	--
49	Aug-Nov	PH6338/ PH4368	Advanced functional Materials	2	16	--



New initiatives in teaching

- Assignments based on hands-on experiments or computer-based modelling
 - EP3230/EP36030: Solid State Physics: Students developed an interactive computer-based model to enhance understanding of lattice vibrations, phase and group velocities, dispersion curves, crystal structures of various unit cells, and the conversion from direct lattice to reciprocal lattice.
 - PH5350: Experimental Techniques: Students were tasked with designing a low-cost, home-built experimental setup. They procured small components and utilized our research and teaching labs to develop fundamental physics experiments and instruments, including the Halbach array, lock-in amplifier, and vibrating sample magnetometer.
- Video content for the PH1031: Physics lab (common for 1st year engineering students)
 - During the COVID period, our first-year common Physics lab transitioned to an online format. As the course coordinator, I collaborated with the technical staff to record videos for all experiments, including demonstrations, data collection, and analysis.

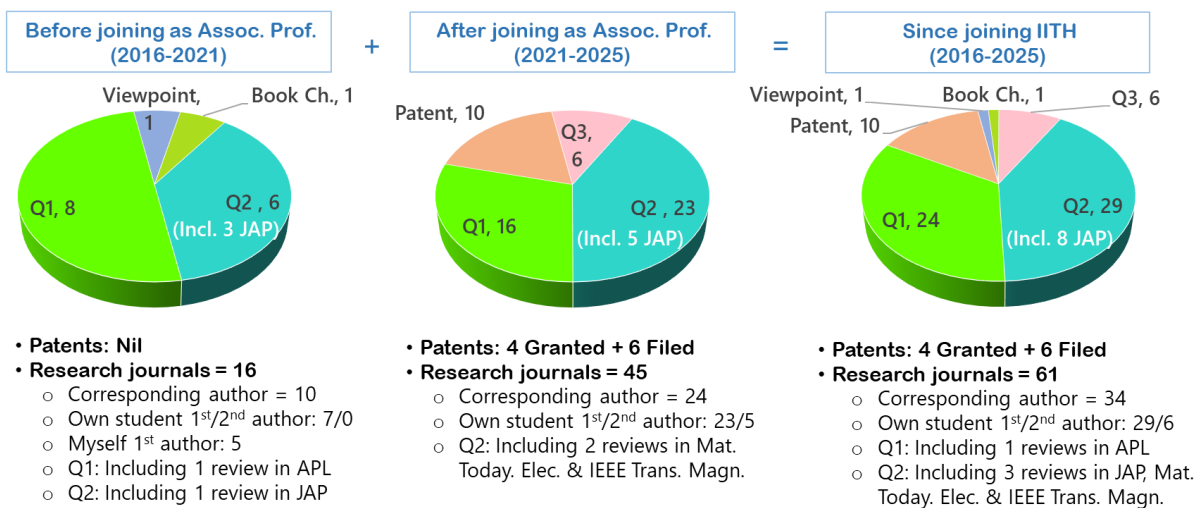
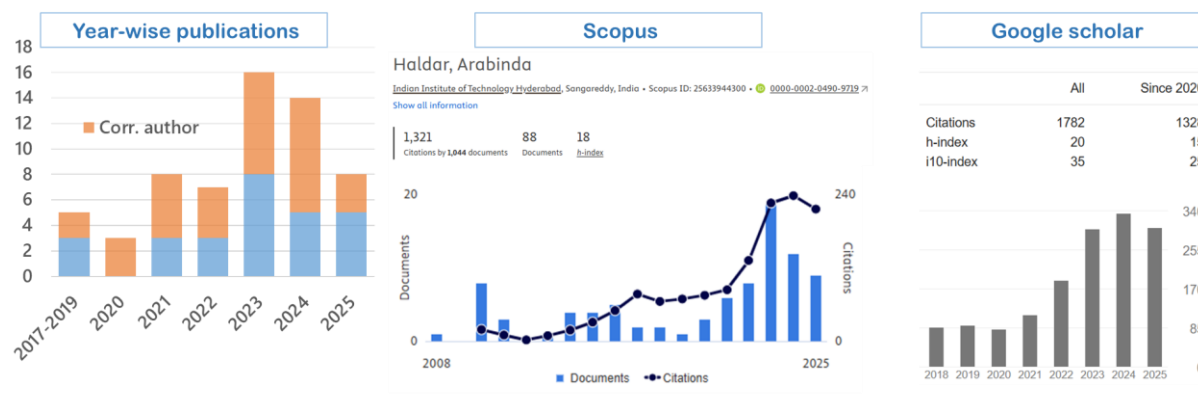
SPONSORED PROJECTS

Sl. No.	Role	Project Title	Sponsoring Agency	Period of project	Project Cost (in Rs., in Lakhs)	Status
1.	PI	Smart magnetic media for isotropic spin wave propagation	ECR-SERB (ANRF)	2018-2021 (3Y)	50.45	Completed
2.	PI	Magnetization dynamics in nanostructures	Seed grant, IIT Hyderabad	2017-2019 (2Y)	3	Completed
3.	PI	Ultrafast magnetization dynamics in magnetic thin film nanostructures	Ramanujan fellowship-ANRF	2017-2022 (5Y)	38	Completed
4.	PI	Control of magnetic microwave properties using external electric current	YSRA-BRNS	2021-2024 (3Y)	25	Completed
5.	PI	Development of self-biased magnetic materials for low loss bias-free passive microwave devices	ARDB-DRDO	2021-2025 (3Y+)	51.2	Completed
6.	Co-PI	Harnessing pure spin current by tailoring molecular spinterface	BRNS	2022-2025 (3Y)	30.32	Completed
7.	Co-PI	Spintronics based Digital Logic Architecture Design for AI Applications	CRG-ANRF	2023-2026 (3Y)	60	Ongoing

8.	PI	Spin wave dispersions and nanoscale imaging of magnons using Brillouin light scattering spectro-microscopy	CRG-ANRF	2023 – 2026 (3Y)	49.5	Ongoing
9.	Co-PI	CISCOM: Centre for in-situ and correlative microscopy (Section: Microscopy of magnetic thin films using MOKE)	DST-SATHI	2024 - 2029	Part of Research Centre	Ongoing

PUBLICATIONS

--SUMMARY--



[Note: JAP is a mainstream journal in spintronics favored by peers.]

*JAP = Journal of Applied Physics (AIP publishing)

-- FULL LIST --

-- 2025 --

87. M. S. Devapriya, N. S. Adithya, A. O. Adeyeye, and A. Haldar*, Controlling different nonlinear dynamics by tailoring magnonic bands in permalloy microconduits, **Physical Review Applied** 24, 024018 (2025). [URL: <https://doi.org/doi.org/10.1103/hmtn-pv2j>]
86. T. Manoj, V. Mattela, A. K. Salimath, S. Debroy, T. R. Kakinada, M. S. Devapriya, A. Haldar, Z. Wen, H. Sukegawa, S. Mitani, C. Murapaka, "Thickness dependent spin to charge interconversion efficiency in polycrystalline BiSb layers deposited on Si substrate", **J. Appl. Phys.** 138, 053903 (2025). [URL: <https://doi.org/10.1063/5.0280905>]
85. R. Jaiswal, M. Kuchibhotla, V. Haragopal, A. Haldar*, C. Murapaka and V. Kannan, "Effect of antidote on magnetization dynamics of Permalloy nanowires", *Physica Scripta*, 100, 095503 (2025). [URL: <https://10.1088/1402-4896/adfb0a>]
84. B. Panigrahi, R. Mondal, C. Murapaka, and A. Haldar*, "Dual Mode Spin-To-Charge Conversion In Ta/NiFe/Ta/CoFeB/Ta Multilayer Thin Films With Varying Spacer Layer Thickness", *IEEE Trans. Magn.* 61, 1300808 (2025).[URL: <https://doi.org/10.1109/TMAG.2025.3591246>]
83. L. A. Longchar, M. Rahaman, M. M. Raja, A. Haldar, Pardeep, G. A. Basheed, S. N. Kaul and S. Srinath, "Effect of the substrate temperature on the Gilbert damping and two-magnon scattering contributions to ferromagnetic resonance linewidth in Co₂FeAl_{0.5}Si_{0.5} Heusler alloy thin films", *J. Phys.: Condens. Matter* 37, 265806 (2025). [URL: <https://doi.org/10.1088/1361-648X/ade651>]
82. K. R. Sahoo, M. Talluri, D. Maity, S. Mundlia, A. Lal, M. S. Devapriya, A. Haldar, C. Murapaka, T. N. Narayanan, "Vanadium Doped Magnetic MoS₂ Monolayers of Improved Electrical Conductivity as Spin-Orbit Torque Layer", **Adv. Funct. Mater.** 2502408 (2025). [URL: <https://doi.org/10.1002/adfm.202502408>]
81. R. Singh, R. Kumar, B. Panigrahi, A. Haldar, P. Kumar, "Magnetic field-driven charge carrier dynamics and time-reversal symmetry breaking in individual TI/FM heterostructures (Bi₂Se₃/Ni₈₀Fe₂₀/p-Si, Bi₂Te₂Se/Ni₈₀Fe₂₀/p-Si, and TlBiSe₂/Ni₈₀Fe₂₀/p-Si) via pump-probe spectroscopy", *Journal of Physics and Chemistry of Solids* 200, 112594 (2025). [URL: <https://doi.org/10.1016/j.jpics.2025.112594>]
80. R. Mondal, V. Alman, A. Haldar and C. Murapaka, "Impact of Cr insertion layer on spin transport properties in Co₂FeAl/Ta system", *J Supercond. Nov. Magn.* 38, 76 (2025). [URL: <https://doi.org/10.1007/s10948-025-06914-x>]

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79. J. Pradhan; M. S. Devapriya, R. Mondal, J. Uzuhashi, T. Ohkubo, S. Kasai, C. Murapaka, and A. Haldar*, "Ultra-low Gilbert damping and self-induced inverse spin Hall effect in GdFeCo thin films." **J Appl. Phys.** 136, 203903 (2024). [URL: <https://doi.org/10.1063/5.0231132>]

78. M S Devapriya, S N. Aditya, M. Kuchibhotla, A. O. Adeyeye, A. Haldar*, Effect of Width and Thickness on Propagating Spin Waves in Permalloy Microstripe Waveguides, **J Appl. Phys.** 136, 093905 (2024). [URL: <https://doi.org/10.1063/5.0223672>]
77. B. Panigrahi, M. M. Raja, C. Murapaka, A. Haldar*, "Spin-to-charge conversion via dual-mode ferromagnetic resonance in Ta/NiFe/FeMn/CoFeB multilayer", **J Magn. Magn. Mater.** 608, 172420 (2024). [URL: <https://doi.org/10.1016/j.jmmm.2024.172420>]
76. M. Kuchibhotla, A. Haldar* and A. O. Adeyeye, "Magnetization dynamics in single and trilayer nanowires", **J Phys: Cond. Mat.** 36, 455808 (2024). [URL: doi.org/10.1088/1361-648X/ad6c9a]
75. S. Sivasubramani, B. Paikaray, M. Kuchibhotla, A. Haldar, C. Murapaka, and A. Acharyya, "Area Efficient Skyrmion Logic based Approximate Adder Architecture Design Methodology", *Transactions on Emerging Topics in Computing*, (Accepted, July 2024). [URL: <https://doi.org/10.1109/TETC.2024.3434723>]
74. S. Singh, D. A. Babu, D. M. Rajkumar, A. K. Nayak, A. Haldar* and M. M. Raja, "Microstructural and magnetic properties of nanocrystalline Nd-Fe-B rare earth magnet prepared by spark plasma sintering technique", **J Supercond. Nov. Magn.** 37, 1689-1700 (2024). [URL: <https://doi.org/10.1007/s10948-024-06806-6>]
73. K. Sriram, Y. S. Pappu, V. C. Peddiraju, R. Mondal, M S Devapriya, A. Haldar, C. Murapaka, "Deposition pressure controlled phase tailoring and stability of β -W for spintronic applications", **J Appl. Phys.** 136, 045302 (2024). [URL: <https://doi.org/10.1063/5.0202304>]
72. M S Devapriya, X. Zhou, A. Haldar*, A. O. Adeyeye, "Spin Textures in High-Aspect-Ratio Ni₈₀Fe₂₀ Nanodisk Arrays: Implications for Next-Generation Spintronic Devices" **ACS Appl. Nano Mater.** 7, 15096-15103 (2024). [URL: <https://doi.org/10.1021/acsanm.4c01857>]
71. R. Singh, S. K. Yadav, R. Kumar, A. Haldar and P. Kumar, "Effect of external/internal magnetic field on photocurrent in Py- Topological insulator heterojunction Ni₈₀Fe₂₀/TI (Bi₂Te₃/Bi₂Se₃/Bi₂Te₂Se)/p-Si devices, *Physical Chemistry Chemical Physics*, 26, 16708 – 16718 (2024). [URL: <https://doi.org/10.1039/D4CP01557F>]
70. B. Panigrahi, M. Manivel Raja, C. Murapaka, A. Haldar*, "Dual mode spin to charge conversion using inverse spin Hall effect in NiFe/FeMn/NiFe multilayer thin films", **J Phys D: Appl. Phys** 57, 305005 (2024). [URL: <https://doi.org/doi.org/10.1088/1361-6463/ad42aa>]
69. M. Kuchibhotla, A. Haldar* and A. O. Adeyeye, "Investigation of magnetization Dynamics in Trilayer Width-modulated Nanowires", **J Appl. Phys.** 135, 103903 (2024). [URL: <https://doi.org/10.1063/5.0191447>]
68. R. Mondal, M S Devapriya, J. Pradhan, A. Haldar, C. Murapaka, "Effect of growth rate on structural, magnetic and spin dynamic properties of Co₂FeAl thin films", **Thin Solid Films** 792, 140268 (2024). [URL: <https://doi.org/10.1016/j.tsf.2024.140268>]

67. K. Sriram, Y. S. Pappu, M S Devapriya, J. Pradhan, A. Haldar, C. Murapaka, "Deposition Pressure Dependence on Spin Hall Angle of W Thin Films Grown on NiFe", SPIN 2340027 (2024). [URL: <https://doi.org/10.1142/S2010324723400271>]

66. S. Sara, C. Murapaka and A. Haldar*, "Voltage-controlled magnetic anisotropy gradient-driven skyrmion-based half-adder and full-adder", *Nanoscale*, 16, 1843-1852 (2024) [URL: <https://doi.org/10.1039/D3NR05545K>].

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65. J. Pradhan, M S Devapriya, R. Mondal, T. Manoj, C. Murapaka, A. Haldar*, Effect of thermal annealing on the magnetization reversal and spin dynamics in ferrimagnetic TbCo thin films, *Journal of Magnetism and Magnetic Materials*, 587, 171363 (2023). [URL: <https://doi.org/10.1016/j.jmmm.2023.171363>]

64. K. Sriram, R. Mondal, J. Pradhan, A. Haldar, C. Murapaka, "Structural Phase Engineering of ($\alpha + \beta$)-W for a Large Spin Hall Angle and Spin Diffusion Length", *Journal of Physical Chemistry C*, 127(46), pp. 22704–22712 (2023). [URL: <https://doi.org/10.1021/acs.jpcc.3c04404>]

63. R. Singh, G. K. Maurya, V. Gautam, R. Kumar, M. Kumar, K. G. Suresh, B. Panigrahi, C. Murapaka, A. Haldar and P. Kumar, "Proximity induced band gap opening in topological-magnetic heterostructure (Ni80Fe20/p-TiBiSe2/p-Si) under ambient condition", *Sci. Rep.* 13, 22290 (2023). [URL: <https://doi.org/10.1038/s41598-023-49004-5>]

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61. K. Sriram, J. Pradhan, M. S. Devapriya, A. Haldar* and C. Murapaka, "Annealing dependence on magnetization dynamics and two-magnon scattering in Co40Fe40B20 thin films", *Thin Solid Films* 779, 139924 (2023). [URL: <https://doi.org/10.1016/j.tsf.2023.139924>]

60. B. Panigrahi, M. Manivel Raja, Jaivardhan Sinha, C. Murapaka, A. Haldar*, "NiFe/FeMn exchange biased systems for bias-field-free magnetization dynamics", *Thin Solid Films* 779, 139923 (2023). [URL: <https://doi.org/10.1016/j.tsf.2023.139923>]

59. R. Gupta, J. Pradhan, A. Haldar, C. Murapaka and P.C. Mondal, "Chemical approach towards broadband spintronics on nanoscale Pyrene films", *Angewandte Chemie* e202307458 (2023). [URL: <https://doi.org/10.1002/anie.202307458>]

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57. M. Kuchibhotla, A. Haldar* and A. O. Adeyeye, "Field angle dependent resonant dynamics of artificial spin ice lattices", *Nanotechnology* 34, 325302 (2023). [URL: <https://doi.org/10.1088/1361-6528/acd2e2>]

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53. S. Singh, A. B. Diraviam, R. Sarkar, A. Haldar*, and M. R. Muthuvel, "Effect of quenching rate on the structural and hard magnetic properties of Nd-Fe-B melt-spun ribbons", **AIP Advances** 13, 025301 (2023). [URL: <https://doi.org/10.1063/9.0000524>]
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49. A. Haldar*, "Functional nanostructures for bias-magnet-free and reconfigurable microwave magnetic devices", **Materials Today Electronics** 2, 100008 (2022). [URL: <https://doi.org/10.1016/j.mtelec.2022.100008>]

[AH was invited to write this review article]

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46. B. Panigrahi, S. K. Sahoo, Syamlal S. k., J. Sinha, H. Basumatary, M. M. Raja and A. Haldar*, "Effect of Ta capping layer on spin dynamics in Co₅₀Fe₅₀ thin films" **Solid State**

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44. B. Paikaray, M. Kuchibhotla, A. Haldar and C. Murapaka, "Reconfigurable logic operations via gate controlled skyrmion motion in a nanomagnetic device", ACS Applied Electronic Materials, 4, 2290 (2022). [URL: <https://doi.org/10.1021/acsaelm.2c00122>]
43. A. Chumak....., A. Haldar,..... et al., "Roadmap on Spin-Wave Computing", IEEE Trans. Magn. 58, 0800172 (2022). [URL: <https://doi.org/10.1109/TMAG.2022.3149664>]

[This is a review article]

-- Before joining as Associate Professor --

-- 2021 --

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41. K. Mahathi, A. Talapatra, A. Haldar* and A. O. Adeyeye, "Magnetization dynamics of single and tri-layer nanodots" J. Appl. Phys. 130, 083906 (2021). [URL: <https://doi.org/10.1063/5.0060689>]
40. A. Haldar* and A. O. Adeyeye, "Functional Magnetic Waveguides for Magnonics", Appl. Phys. Lett. 119, 060501 (2021). [URL: <https://doi.org/10.1063/5.0061528>]

[This is an invited 'perspective' article]

39. B. Paikaray, A. Joseph, C. Murapaka and A. Haldar*, "Tunable microwave properties of a skyrmion in an isolated nanodisk", J Magn. Magn. Mater. 529, 167900 (2021). [URL: <https://doi.org/10.1016/j.jmmm.2021.167900>].
38. B. Paikaray, M. Kuchibhotla, C. Murapaka and A. Haldar*, "Skyrmion dynamics in concentric and eccentric nano-ring structures", IEEE Trans. Magn. 58, 4300406 (2021). [URL: <https://doi.org/10.1109/TMAG.2021.3086487>]
37. T. Manoj, S. Kotha, B. Paikaray, D. Srideep, A. Haldar, K. V. Rao and C. Murapaka, "Giant spin pumping at ferromagnet (Permalloy) - organic semiconductor (perylene diimide) interface", RSC Advances 11, 35567 (2021) [URL: <https://doi.org/10.1039/D1RA07349D>]
36. K. Sriram, J. Pala, B. Paikaray, A. Haldar and C. Murapaka, "Effect of seed layer thickness on Ta crystalline phase and spin Hall angle", Nanoscale 13, 19985 (2021). [URL: <https://doi.org/10.1039/D1NR06007D>]

35. A. Haldar and A. Barman, "A new drift in spin-based electronics", Physics (APS publishing) 14, 92 (2021). [URL: <https://physics.aps.org/articles/v14/92>]

[It is an invited "viewpoint" article (commentaries by experts)]

-- 2020 --

34. A. Haldar* and A. O. Adeyeye, "Reconfigurable and self-biased magnonic meta-materials", J. Appl. Phys. 128, 240902 (2020). [URL: <https://doi.org/10.1063/5.0033254>].

Highlights: This article has been "featured" in J. Appl. Phys.

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33. K. Begari and A. Haldar*, "Reconfigurable microwave properties of zigzag magnetic nanowires", J. Phys. D: Appl. Phys. 53, 455005 (2020) [URL: <https://doi.org/10.1088/1361-6463/aba571>].

32. A. Haldar* and A. O. Adeyeye, "Microwave assisted gating of spin wave propagation", Appl. Phys. Lett. 116, 162403 (2020). URL: <https://doi.org/10.1063/5.0006945>

-- 2017 to 2019 --

31. K. Begari and A. Haldar*, "Reconfigurable microwave properties in C-, L- and S-shaped nanomagnets", J. Phys. D: Appl. Phys. 52, 335003 (2019). URL: <https://doi.org/10.1088/1361-6463/ab24a5>

30. A. Haldar and A. O. Adeyeye, "Reconfigurable magnetic and microwave properties of a ferrimagnetic-type artificial crystal", J. Appl. Phys. 123, 243901 (2018). URL: <https://doi.org/10.1063/1.5032158>

News: "A new ferrimagnetic-type magnetic metamaterial improves energy efficiency" [URL: <https://doi.org/10.1063/1.5045477>]

Highlights: Our work is featured on the cover of J. Appl. Phys. ([Click here](#)).

29. K. Begari and A. Haldar*, "Bias-free giant tunability of microwave properties in multilayer rhomboid nanomagnets", J. Phys. D: Appl. Phys. 51, 275004 (2018). URL: <https://doi.org/10.1088/1361-6463/aac86a>

28. Book chapter: G. Shimon, A. Haldar, and A. O. Adeyeye, "Magnetization dynamics of reconfigurable 2D magnonic crystals", Book: Spin wave confinement: Propagating spin waves, Second Edition, Pan Stanford Publishing (2017), Edited by Sergej O. Demokritov. URL: <http://dx.doi.org/10.1201/9781315110820-5>

27. A. Haldar, C. Tian and A. O. Adeyeye, "Isotropic transmission of magnon spin information without a magnetic field", Science Advances, 3, e1700638 (2017). URL: <http://10.1126/sciadv.1700638>

News: "Engineers achieve significant breakthrough in spin wave based information processing technology". URL:

<https://scienceadvances.altmetric.com/details/22261920/news>

-- Before joining IIT Hyderabad --

26. D. Kumar, P. Lupo, A. Haldar and A. O. Adeyeye, "Unconventional spin distribution in thick $Ni_{80}Fe_{20}$ nanodisks," Appl. Phys. Lett. 108, 192404 (2016).
25. A. Haldar and A. O. Adeyeye, "Bias field free tunability of microwave properties based on geometrically controlled Permalloy nanomagnets," Appl. Phys. Lett. 108, 162401 (2016).
24. A. Haldar, D. Kumar, and A. O. Adeyeye, "A reconfigurable waveguide for energy-efficient transmission and local manipulation of information in a nanomagnetic device," Nature Nanotechnology, 11, 437-443 (2016). URL: <https://doi.org/10.1038/nnano.2015.332>

News and Highlights:

- **APS Research News - Editor's Choice:** "Channeling Spin waves", Physics 9, 73 (June 30, 2016). URL: <http://physics.aps.org/articles/v9/73>
 - **Nature Nanotechnology highlight:** This work has been selected for TOC image of Nature Nanotechnology (May issue). URL: <http://www.nature.com/nnano/journal/v11/n5/pdf/toc.pdf>
 - **Nature news:** "Spintronics: Nanomagnonics around the corner" by D. Grundler. URL: <http://dx.doi.org/10.1038/nnano.2016.16>
 - **University news:** "NUS researchers developed energy efficient information processing for spintronic devices", URL: <https://www.ece.nus.edu.sg/home/news.html>
23. A. Haldar and A. O. Adeyeye, "Deterministic control of magnetization dynamics in reconfigurable nanomagnetic networks for logic applications," ACS Nano 10, 1690-1698 (2016). URL: <http://dx.doi.org/10.1021/acsnano.5b07849>
 22. A. Haldar and A. O. Adeyeye, "Artificial metamaterials for reprogrammable magnetic and microwave properties," Appl. Phys. Lett. 108, 022405 (2016).
 21. G. Riley, H. J. Liu, M. Asmat-Uceda, A. Haldar, and K. S. Buchanan, "Observation of the dynamic modes of a magnetic antivortex using Brillouin light scattering", Phys. Rev. B 92, 064423 (2015).
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18. A. Ganguly, R. Robinson, A. Haldar, J. Sinha, A. T. Hindmarch, D. A. Atkinson, A. Barman, "Time-domain detection of modulation of damping and spin Hall angle in Pt/Ni₈₁Fe₁₉ bilayer film." Appl. Phys. Lett. 105, 112409 (2014).
17. **Book chapter:** "Time domain study of magnetization dynamics in magnetic thin films, micro- and nanostructures", A. Barman and A. Haldar, *Solid State Physics (Book series)*, Vol. 65, 2014, pp. 1-108. Burlington: Academic Press, Editors: Robert E. Camley and Robert L. Stamps. [URL: <http://dx.doi.org/10.1016/B978-0-12-800175-2.00001-7>]
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14. A. Haldar and Kristen S. Buchanan, "Magnetic antivortex formation in pound-key-like nanostructures", Appl. Phys. Lett. 102, 112401 (2013).
13. A. Haldar, A. Das, A. Hoser, T. Hofmann, A. K. Nayak, K. G. Suresh and A. K. Nigam. "High field neutron diffraction studies in Ce(Fe_{0.95}Si_{0.05})₂ compound." J. of Appl. Phys. 109, 07E125 (2011).
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10. A. Haldar, K. G. Suresh and A. K. Nigam. "Observation of re-entrant spin glass behavior in Er doped CeFe₂ compound." Europhys. Lett. 91, 67006 (2010).
9. A. Haldar, N. K. Singh, Ya. Mudryk, K. G. Suresh, A. K. Nigam, and V. K. Pecharsky. "Temperature and magnetic field induced structural transformation in Si doped CeFe₂: An in-field X-ray diffraction study." Solid State Commun. 150, 879-883 (2010).
8. A. Haldar, N. K. Singh, Ya. Mudryk, K. G. Suresh, A. K. Nigam, and V. K. Pecharsky. "Magnetostructural transition in Ce(Fe_{0.975}Ga_{0.025})₂ compound." J. of Appl. Phys. 107, 09E133 (2010).
7. A. Haldar, K. G. Suresh and A. K. Nigam, "Magnetization jumps and relaxation in doped CeFe₂." J. of Phys.: Conf. Series 200, 032021 (2010).
6. A. Haldar, K. G. Suresh and A. K. Nigam. "Martensitic features in Si doped CeFe₂ revealed by magnetization and transport study." Intermetallics 18, 1772-1778 (2010).
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4. A. Haldar, K. G. Suresh, A. K. Nigam. "Magnetization jumps and transport properties of $R_6\text{Co}_{1.67}\text{Si}_3$ ($R=\text{Tb}$ & Nd) ternary alloys." Physica B 405, 3446–3451 (2010).
3. P. Kumar, Niraj K. Singh, A. K. Nayak, A. Haldar, K. G. Suresh and A. K. Nigam "Large reversible magnetocaloric effect in Er_3Co compound." J. of Appl. Phys. 107, 09A932 (2010).
2. A. Haldar, K. G. Suresh, A. K. Nigam, A. A. Coelho and S. Gama. "Stabilization of antiferromagnetism in CeFe_2 alloys: Effects of chemical and hydrostatic pressure." J. Phys.: Condens. Matter 21, 496003 (2009).
1. A. Haldar, K. G. Suresh and A. K. Nigam. "Magnetism in gallium doped CeFe_2 : Martensitic scenario." Phys. Rev. B 78, 144429 (2008).

PATENTS (since joining IITH)

-- Granted --

10. S. Santosh, C. Kishore, A. Haldar, C. Murapaka and A. Acharyya, "System and method for a skyrmionic logical adder for electronic circuit", Indian Patent no.: 569524 (202441031123), **Grant date: 1-Aug-2025**.
9. C. Kishore, S. Sara, S. Santosh, C. Murapaka, A. Haldar and A. Acharyya, "Method and system to operate tsetlin machines using skyrmion", Indian Patent application no.: 202441071705 **Grant date: 21-May-2025**.
8. B. Paikaray, K. Mahathi, A. Haldar, C. Murapaka, "Skyrmion based majority logic gate in a nanomagnetic device", Indian Patent No: 435541 (202241010372), **Grant date: 26-June-2023**.
7. S. Sara, C. Murapaka and A. Haldar, "A skyrmion based device for performing logical operations", Indian Patent no. 559670 (202441036681), **Grant date: 6-Feb-2025**.

-- Filed --

6. S. Sara, S Sivasubramani, C. Kishore, A. Acharya, Chandrasekhar Murapaka, and A. Haldar, "A skyrmion based logic operating circuit", Indian Patent application no.: 202541084119, Filing date: Sep 2025.
5. C. Kishore, S. Santosh, C. Murapaka, A. Haldar and A. Acharyya, "A skyrmion-based tsetlin machine inference system", Indian Patent application no.: 202441086334 Filing date: 9-Nov-2024.
4. C. Kishore, S. Sara, S. Santosh, C. Murapaka, A. Haldar and A. Acharyya, "A Skyrmion based counter", Indian Patent application no.: 202441071776 dated 23-Sep-2024.
3. S. Sivasubramani, A. Haldar, C. Murapaka, and A. Acharyya, "System And Method For Skyrmion Based 3D Low Complex Runtime Reconfigurable Architecture Design Methodology of Universal Logic Gate", Indian Patent application no.: 202341039644, Filed in 2024.

2. B. Panigrahi, A. Haldar, C. Murapaka, "NiFe/FeMn exchange biased systems for bias-field-free magnetization dynamics", Indian Patent Application 202241051146, TEMP/E-1/58726/2022-CHE (Filed Sep 2022).
1. B. Paikaray, K. Mahathi, A. Haldar, C. Murapaka, "Reconfigurable logic via gate controlled skyrmion motion", Indian Patent Application 202141057701, TEMP/E-1/65140/2021-CHE (Filed Dec 2021).

CONFERENCE SUMMARY (since joining IITH)

- 📌 International conference (poster/oral): 35
- 📌 National conference (poster/oral): 3

SEMINARS/INVITED TALKS (since joining IITH)

1. (Invited talk): "Multi-mode ferromagnetic resonance and spin-to-charge conversion in HM/FM/AFM-based thin films", International conference on magnetic materials and applications **ICMAGMA 2025**, Feb 12-14, 2025, IISc Bangalore, India.
2. (Invited talk): "Design of skyrmion-based logic architectures using micromagnetic simulations", **Indo-German Kick-off Workshop** on Recent progress in topological magnetism, Nov 5-8, 2024, NISER/Puri, India.
3. (Invited talk): "Spin wave spectro-microscopy using Brillouin light scattering, July 18-20, 2024, Symposium on magnetism and spintronics (**SMS2024**)", IIT Bombay, India.
4. (Invited talk): "Reconfigurable spin dynamics in self-biased magnetic thin films and nanostructures", Nov 27-30, 2023, Quantum Condensed matter (**QMAT**), NISER, Bhubaneswar.
5. (Invited talk): "Tunable Spin Dynamics in Novel Magnetic Thin Films and Nanostructures", July 18-20, 2023, Quantum Matter Heterostructures (**QMH**), IIT Hyderabad.
6. (Invited talk): "Reconfigurable magnonics using self-biased reprogrammable magnetic structures", Mar 27-29, 2023, **ICONN 2023** (virtual conference), SRM Institute of Science and Technology, Chennai.
7. (Invited talk): "Reconfigurable magnonics using self-biased reprogrammable magnetic structures", Mar 3-4, 2023, **Frontiers in Physics**, University of Hyderabad, India.
8. (FDP-talk): "Fundamentals of Spintronics materials and devices", July 26, 2022, Five Days **Faculty Development Program** on Recent Trends in Condensed Matter Physics, Department of Physics, Amity University, Kolkata.
9. (Invited talk): "Magnon spintronics", Sep 15, 2021, Physics Seminar, **IIT-ISM Dhanbad**, India.
10. (Invited talk): "Reconfigurable magnonics using self-biased reprogrammable nanomagnetic structures", **INTERMAG 2021** (virtual conference), 26-30 April 2021 (was due to be held in Lyon, France), sponsored by the IEEE Magnetics Society.
11. (Invited talk): "Self-biased nanomagnets for energy-efficient magnonic application", 3rd International conference on current trends in materials science and engineering (**CTMSE 2021** – virtual conference), 11-13 March 2021.
12. (Seminar-web): "Magnon spintronics", **W2S seminar**, Sep 3, 2020, NISER, Bhubaneswar, India.
13. (Seminar-web): "Magnon spintronics", June 8, 2020, **Physics Seminar**, IIT Bombay, India.
14. (Invited talk) "*Bias-field free magnetic waveguides for magnonic devices*", 63rd DAE Solid State Physics Symposium (**SSPS**), Dec 18-22, 2018, Hisar Haryana, India.
15. (Invited talk) "*Reprogrammable and energy-efficient magnonic devices*", 4th IEEE-ICEE 2018 International conference on emerging electronics, Dec 16-19, 2018, **IISc Bangalore**, India.

16. (Invited talk) *"Reconfigurable and bias-field-free magnetic waveguides for channeling spin waves"*, International conference on complex and functional materials (ICCFM), Dec 13-16, 2018, **S. N. Bose National Centre** for Basic Sciences, Kolkata, India.
17. (Invited talk) *"Bias-field free magnonic devices"*, International conference on magnetic materials and applications (**ICMAGMA**), Dec 9-13, 2018, NISER, Bhubaneswar, India.
18. (Invited talk) *"Magnon Spintronics"*, Workshop on "Functional Magnetic Materials", Feb 26 – Mar 3, 2018, **University of Hyderabad**, India.
- # Note: AH was also invited to **Magnonics 2019**, July 28- Aug 1, 2019, Italy. However *VISA was not approved*.

PH.D STUDENT SUPERVISIONS

S. No.	Student Name	Fellowship	Joining date	Thesis title	Status
1.	Dr. B. Krishna (ph17resch01002)	CSIR	Dec 2016	Tunable magnetization dynamics in novel thin film nanostructures	Graduated Defense date: 27-Sep-2021 Current position: Technical Staff/Reseracher at Mahindra University
2.	Dr. Brahmaranjan Panigrahi (ph19resch02004)	DST-INSPIRE	Aug 2019	Control of ferromagnetic resonance properties via spin pumping and exchange bias	Graduated Defense date: 1-May-2025 Current position: Appearing for postdoc interview
3.	Dr. Kuchibhotla Mahathi (ph19resch11002)	MoE	Aug 2019	Magnetization dynamics in novel magnetic nanostructures	Graduated Defense date: 17-Sep-2024 Current position: Post-doc at Clarke University, MA, USA
4.	Dr. Bibekananda Paikaray (ms19resch12001) Joint supervision: Dr. C. Murapaka, MSME, IITH	DST-INSPIRE	Aug 2019	Investigation of exotic spin configurations and their dynamics in multilayer magnetic thin film stacks	Graduated Defense date: 21-Feb-2023 Current position: Postdoc at Laboratoire Albert Fert, CNRS, Thales, Univ. Paris-Saclay

5.	Dr. Sudeep Singh (ph20resch13001) Joint supervision: Dr. M. M. Raja, DMRL	MoE (IITH- DRDO)	Sep 2020	Investigation on Nd-Fe-B based permanent magnets	Graduated Defense date: 13-Aug- 2025 Current position: Received a postdoc offer from AIST, Japan
6.	Ms. Devapriya M. S. (ph20resch11005)	MoE	Sep 2020	Spin dynamics in magnetic thin films and nanostructures	Ongoing (OC will be conducted by Nov 2025) Current position: Received a postdoc offer from Tohoku University, Japan
7.	Mr. Jhantu Pradhan (ph21resch01006) Joint supervision: Dr. C. Murapaka, MSME, IITH	CSIR	Dec 2020	Spin dynamics in ferrimagnetic systems	Ongoing (Thesis writing stage)
8.	Mrs. Sarwath Sara (ph21resch01008)	CSIR	Dec 2020	Skyrmion-based logic devices	Ongoing (Thesis writing stage)
9.	Ms. Adithya S Nair (ph23resch11002)	DST- INSPIRE	July 2023	Spin torque effects in spintronic devices	Ongoing
10.	Mr. Ayush Rastogi (id23resch11003) Joint supervision: Prof. S. G. Singh, EE, IITH	MoE (ID program)	Aug 2023	Microwave properties in low damping magnetic ferrite films	Ongoing
11.	Mr. Ambarish Kumar (ph24resch01002)	MoE	Dec 2023	Thin film heterostructures for Dzyloshinskii Moriya interactions	Ongoing
12.	Mr. Sourav Rajat Subhra Maitra (ph24resch01007)	MoE	Dec 2023	Spin torque effect in topological insulators	Ongoing

13.	Ms. Srilalitha Gajarao (id23m25p100001) Joint supervision: Dr. C. Murapaka, MSME, IITH	MoE (ID program)	Aug 2025	Skyrmion based logic architectures	Ongoing
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INFRASTRUCTURE ESTABLISHMENT AT IITH

I would like to emphasize that we developed/assembled most of our experiments in-house. We meticulously designed and procured the components, followed by their assembly and standardization. This process of instrumentation is inherently challenging, requiring continuous maintenance and troubleshooting, including tasks such as optical alignment and signal-to-noise ratio (SNR) monitoring. We are truly grateful to the institute and administration for their unwavering support; without it, we would not have been able to transform our lab into a renowned center for magnonic research globally. Below are some key achievements that have emerged from our dedicated efforts over the years.

- **Broadband FMR: Ferromagnetic resonance** (Funding: project: ECR-ANRF):
 - Procured and commissioned room-temperature (RT) FMR system: Frequency: 2-18 GHz, Field: +/- 3500 Oe
 - Development of a custom-built sample stage for different measurement capabilities using our workshop
 - In-house integration of the **inverse spin Hall effect (ISHE)** experiment
 - Developed MATLAB code for batch processing and data analysis
 - Since commissioning in 2019: 50+ publications, several collaborations
- **Home-built Micro-BLS: Micro-focused Brillouin light scattering** (Funding: JICA, CRG-SERB, DRDO, and IITH): BLS microscopy represents cutting-edge technology in the field of nano-spintronics, offering a unique capability to investigate high-frequency magnons—quanta of spin waves - up to 500 GHz. Among its various modes of operation, BLS microscopy facilitates nanoscale imaging of magnons. There are only a few such microscopy systems available worldwide, and the **BLS setup at IITH is the first and only magnon microscopy facility working in India.**
 - Custom-built and assembled system by procuring several instruments via 20+ POs
 - Designed custom components (~ 50) for the setup using AutoCAD and CNC-fabricated at the IITH workshop and outside.
 - Designed an economic darkroom/cleanroom setup using a local vendor
 - Since commissioning in 2024: 2 publications, 2 submitted, and several collaboration interests globally
- **Home-built: ST-FMR: Spin torque Ferromagnetic resonance** (Funding: project: ARDB-DRDO):
 - Designed and custom-built by assembling different components.
 - Developed LabVIEW/MATLAB code for performing experiments and data analysis
 - Only a handful of such facilities are available in India. Frequency: 0-24 GHz
 - Since commissioning in 2024: 1 publication in Adv Funct. Mater.

- **Sputtering: Thin film deposition** (Funding: project: CRG-ANRF):
 - Procured and commissioned a 2 two-chamber UHV system (commercial) through a joint proposal
 - Suitable for RF and DC magnetron sputtering
- **Cryo-MOKE** (Funding: DST-SATHI, IITH):
 - Procured and commissioned a low-temperature MOKE microscopy system (commercial system)
 - Temperature: 90-400 K, Field: Vector field: +/- 4000 Oe
- **Photolithography** (Funding: project: BRNS, IITH):
 - Procured and commissioned a photolithography system (commercial) via a joint proposal. Resolution: 2 micron
 - Establishment of peripheral equipment: Spin coater, Optical microscope, Hot plate
- **Home-built probe station for magneto-transport** (Funding: project: YSRA-BRNS):
 - Home-built programmable field variation using a permanent magnet for the probe station
 - Suitable for transport experiments at RT in the presence of an angular field
 - Developed LabVIEW/MATLAB code for performing experiments and data analysis
- **MOKE/Faraday effect** (Funding: IITH):
 - Procured and commissioned a MOKE system (commercial), which is suitable for micron-thick samples.
- **Home-built masks and coplanar waveguides** (Funding: project: ANRF, DRDO):
 - One UV mask was fabricated at Cense, IISc Bangalore. The second UV mask was fabricated commercially by a company based in Singapore.
 - Our own chip design with 36 coplanar waveguides (CPWs) are prepared using the cleanroom facility and with a maskless lithography system.
 - We have designed 50 ohm impedance stand-alone CPWs suitable for frequencies up to 50 GHz. Our CPW performance is comparable to that of commercial ones, but with an order of magnitude lower cost.
- **Home-built VSM: Vibrating sample magnetometer** (Funding: project: YSRA-BRNS):
 - An economic VSM was home-built by fabricating our own 2D planar pick-up coils
 - Lock-in-based measurements along with a regular vibrator.
 - Developed LabVIEW code for performing experiments and data analysis
 - The system is capable of measuring thin films and compares well with a commercial system.
 - MSc students and project interns built this tool.
- **Micromagnetic simulations** (Funding: project: Ramanujan Fellowship, ANRF):
Established infrastructure for micromagnetic simulations, both CPU (OOMMF) and GPU (Mumax) based.
 - Development of codes for data analysis and post-processing
 - One CPU-based workstation (Intel Xeon, 10 cores, 64GB RAM)
 - 2 GPU-based workstations (NVidia 11Gb, 4352 cores)

- **Under construction**
 - **Low-cost maskless lithography** (Funding: RDF, project: CRG-ANRF):
 - An economic maskless lithography system is being developed using a projector and microscope as part of an MSc project. This will be useful for creating contact pads with dimensions in the 10's of microns.
 - **Quadrupole magnet** (Funding: RDF):
 - We are building a custom-designed quadrupole magnet as part of an MSc project. This will be useful for creating an angular magnetic field with ample optical/probe access.

ADMINISTRATIVE ACTIVITIES

- Member of DUGC, PhD Admission committee, Faculty search committee, Dept. vision committee
- FIC for the Builder's lab in the Department of Physics, since 2025.
- Department coordinator for Annual report: 2017- 2020
- Sunshine (counseling service) representative of the department: 2019 – 2023
- Committee Member, Technology Incubation Centre, Start-up India Initiative
- Faculty advisor of EP17 batch.
- IITH Library Committee Member: 2018 - 2023
- Coordinator, PhD Admission to Physics: 2016 - 2018.

OUTREACH

- Reviewer responsibility: **Physical Review Letters**, Physical Review Applied, Nature Communications, Nano Letters, Nanoscale, Scientific Reports, Journal of Applied Physics, Journal of Physics D: Applied Physics, Physica B, Journal of Magnetism and Magnetic Materials, Journal of Physics Condensed Matter, and J. Alloys and Compounds.
- Reviewers for the ANRF, BRNS, and I-HUB proposals (**National**).
- Organized **TEQIP online workshop** on "Magnetic materials for MEMS-based devices" at IIT Hyderabad, Oct 29 – Nov 01, 2020.
- Participated in the **Faculty Development Program**, Department of Physics, Amity University, Kolkata, July 26, 2022.
- Local organizing committee, **iCOLD** conference, held at IIT Hyderabad, March 6-8, 2025.
- Local organizing committee, **QMH** conference held at IIT Hyderabad, July 18 – 20, 2023
- **Convener** of the upcoming Symposium on Magnetism and Spintronics (**SMS-3**) international conference at IIT Hyderabad, to be held from January 21-24, 2026.
- Playing an active role in the ongoing effort to establish collaborative research between IITH and **Durham University**. Both sides (IITH & DU) have recognized the existing fruitful research collaboration between my group and Durham University.

COLLABORATIVE RESEARCH (since joining IITH)

I would like to highlight the significant impact of the collaborative research initiatives facilitated by the institute. The institutional support, through joint programs, student exchange initiatives, and interdisciplinary collaborations, has been vital for our group's exceptional growth over the past few years. This support has shaped our research directions, enabling us to achieve high-quality and

impactful outcomes through collaborative efforts that leverage complementary expertise. Below is a summary of our major collaborative efforts.

- **IITH-NIMS program: Dr. Shinya Kasai**
 - A. Haldar visited NIMS as a guest researcher. Also, one PhD student spent 1 year at NIMS
- **IITH-NTHU program: Prof. Lai**
 - One PhD student is working under this program
- **IITH-DRDO program: Dr. Manivel M. Raja, DMRL**
 - One PhD student has completed his thesis under the PhD program
- **IITH ID PhD program**
 - One student is working under this program along with Prof. Shiv G. Singh, EE
 - One student is working under this program along with Dr. Chandrasekhar Murapaka, MSME
- **Through a sponsored project**
 - NISER: Prof. Subhankar Bedanta: One joint-sponsored project as PI (CRG, ANRF)
 - EE-IITH: Prof. Shiv G. Singh and Dr. Abhishek Kumar: One joint sponsored project as PI (ARDB-DRDO)
 - MSME-IITH: Dr. Chandrasekhar Murapaka: Two joint-sponsored projects as co-PI (CRG-ANRF and BRNS)
 - EE-IITH: Prof. Amit Acharya: One joint-sponsored project as co-PI (CRG-ANRF)
 - CHY, IITH: Dr. Venkata R. Kotagiri: One joint-sponsored project as co-PI (BRNS)
- **Other collaborations**
 - Prof. Adekunle O. Adeyeye, Durham University, UK: Large area lithographic fabrication of nanostructures (wafer-scale) (working on UKERI proposal)
 - Prof. S. N. Piramanayagam (Prem), NTU, Singapore: Domain-wall neuromorphic devices
 - Dr. Jaivardhan Sinha, SRMIST, Chennai: Magnetic characterizations and thin film growth
 - Prof. T. N. Narayanan, TIFR Hyderabad: Spintronics with 2D materials (recently published in Adv. Funct. Mater. (2025))
 - Prof. Karthik Raman, TIFR Hyderabad: Low-temperature magneto-transport measurements (submitted ARG-ANRF 2025 proposal jointly)
 - Dr. Prakash Chandra Mondal, IIT Kanpur: Organic spintronics (recent publication in Angewandte Chemie (2023))
 - Dr. Debanjan Bhowmik, EE, IIT Bombay: Spin-orbit torque devices and micro-BLS (joint proposal submitted under Quantum Mission)
 - Prof. Srinath, UoH: Ferromagnetic resonance
 - Prof. Pramod Kumar, IIIT Allahabad: Topological materials
 - Prof. Ajaya K. Nayak, NISER: Low temperature measurements
 - Dr. Murtaza Bohra, Mahindra University, Hyderabad: Magnetic ferrites
 - Dr. Vijayanandini Kannan, GITAM, Hyderabad: Spin dynamics in nanostructures

ABROAD VISIT

- A. Haldar
 - Sendai, Japan: visited to chair a session for the INETRMAG conference during May 15-19, 2023
 - NIMS, Japan: visited as a guest researcher for two weeks in November 2024
- PhD students
 - Warsaw, Poland: Two PhD students visited to attend the Joint European Magnetism Symposia (JEMS) during July 24-29, 2022
 - Sendai, Japan: Two PhD students visited to attend the INTERMAG conference from May 15-19, 2023
 - NIMS, Japan: One PhD student spent a year under the IITH-NIMS program from April 2024 to April 2025.
 - New Orleans, USA: One PhD student visited to attend the Joint MMM-INTERMAG conference from January 13-17, 2025
 - San Diego, USA: One PhD student visited to attend the IEEE magnetic society summer school from June 22-26, 2025

REFERENCES

1. Prof. K. G. Suresh

Professor, Department of Physics

IIT Bombay, Powai, Mumbai

Mobile: 9869649706

Email: suresh@phy.iith.ac.in

2. Prof. Adekunle Adeyeye, FAPS, FInstP

Principal of Trevelyan College & Professor of Physics

Royal Society Wolfson Fellow

Durham University

Elvet Hill Road, Durham, DH1 3LN

Email: adekunle.o.adeyeye@durham.ac.uk

3. Prof. Subhankar Bedanta

Professor in Physics

National Institute of Science Education and Research (NISER) Bhubaneswar

PO- Jatni, Khurda, Odisha - 752050, India

Phone: +91-674-2494232

Email: sbedanta@niser.ac.in

4. Prof. Paolo Vavassori

IKERBASQUE Professor and co-leader of the Nanomagnetism group

CIC nanoGUNE BRTA, Tolosa Hiribidea, 76

20018 Donostia - San Sebastián, Spain

Phone: +34 943 57 4010

Email: p.vavassori@nanogune.eu